

Mathematical Physics and Computation

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CHAPTER I
Measure Theory

1.1 Measure

It will be inevitable to step on some measure & integration theory in dealing with *Partial Differential Equations* and *Stochastic Processes*. This chapter acts as a brief introduction.

σ -Algebra

Motivation In physics, the task of finiding “volume” is ubiquitous. For instance in computing mass, charge, the integral would resemble:

$$\int (\cdot) dV \tag{1}$$

In some sense, this dV is a function $dV : E \rightarrow \mathbb{R}$ which computes the volume of $E \subset \mathbb{R}^n$, a tiny control volume we are interested in. It would be very natural to ask for this function to satisfy:

CHAPTER II

Linear Operator

- 2.1 Finite-dimensional Vector Space
- 2.2 Infinite-dimensional Vector Space

CHAPTER III
Calculus of Variation

CHAPTER IV

Partial Differential Euqations

4.1 Introduction

4.2 Elementary Methods

CHAPTER V
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CHAPTER VI

Miscellaneous

6.1 Basic Numerics

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